PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: LRT AI Operations - Decision Support System

Version: 1.0

1. Project Overview

- **Problem Statement:** Malaysian LRT duty managers must make rapid scheduling decisions when disruptions occur - sudden weather changes, on-track emergencies, or large public events that drive passenger surges. These decisions are made manually under time pressure, with the duty manager mentally balancing passenger demand, train capacity, weather safety limits, emergency protocols, and operating cost. The process is slow (15–20 minutes per disruption), inconsistent across shifts, and prone to two costly failure modes: over-provisioning trains (wasting RM thousands per day per train-hour) or under-serving demand (overcrowded platforms and stranded passengers). Frontline staff have no unified tool that fuses fragmented operational data with reasoning capability.
- **Target Domain:** AI for Economic Empowerment & Decision Intelligence - applied to public transport operations management. The product targets the operational decision layer of urban rail, where every scheduling choice has direct cost, productivity, and public-service implications.
- **Proposed Solution Summary:** An AI-powered decision-support system in which Z.AI's GLM (ilmu-glm 5.1) is the central reasoning engine. The duty manager picks a date and line, then describes the situation through one of two input channels: a manual form or a free-text description. The system instantly computes three scheduling options: conservative, moderate, and aggressive - each showing predicted train frequency change, load factor, cost delta, and passengers served. GLM then reads the structured situation, the calculated options, and historical precedents from past incidents, and recommends the optimal choice with a plain-language explanation. The duty manager applies the chosen option to the day's schedule with one click. The decision and updated schedule are saved to a database for audit and future retrieval. Without GLM, the system can still calculate the

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three options but cannot interpret messy text/image input, cannot reason across context and history, and cannot explain trade-offs - making GLM essential to the core product value.

2. Background & Business Objective

- **Background of the Problem:** When a disruption occurs today, LRT duty managers consult fragmented sources: hourly ridership tables, weather forecasts, event schedules from external promoters, internal incident logs, and unwritten operational heuristics. They mentally combine these inputs to estimate passenger surge, then pick a frequency adjustment based on experience. Two managers facing the same situation can make different decisions because they weigh the variables differently. Documentation is informal, making it difficult to learn from past incidents, onboard new staff, or audit how a particular shift was handled. For example, as Klang Valley ridership grows and event density increases (concerts, sports finals, festival seasons), the cognitive load on duty managers continues to rise.
- **Importance of Solving This Issue:** Each unnecessary train deployed costs RM 350 per train-hour (example). Across an 18-hour operating day on three lines, even small over-provisioning compounds into tens of thousands of ringgit per week. Conversely, for example, under-provisioning during a Coldplay-scale concert can leave thousands of passengers stranded, eroding public trust in rail as the preferred transit option. By compressing the decision cycle from 15–20 minutes of manual analysis to under 30 seconds of AI-assisted review, the system unlocks both cost reduction and service-quality improvements simultaneously. This is a textbook economic-empowerment outcome: a public service that delivers more value at lower cost, without requiring frontline staff to carry the analytical burden alone.

- **Strategic Fit / Impact:** The solution aligns with Malaysia's push toward smart public infrastructure and the use of AI to lift productivity in government-linked services. It demonstrates how Z.AI's GLM can be embedded in operational workflows at the decision

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layer, not just the chatbot layer - reasoning across structured ridership data, unstructured situation descriptions, and historical context. Beyond LRT, the same architecture (rule-based math layer + GLM reasoning layer + persistence) generalizes to bus networks, ride-hailing fleet balancing, hospital staff rostering, and any operational domain where frontline staff must trade off cost, capacity, and safety in real time.

3. Product Purpose

3.1. Main Goal of the System: To give LRT duty managers an AI-powered decision partner that turns fragmented operational data into a clear, justified scheduling recommendation in seconds by reducing operating cost, improving passenger service, and standardizing decision quality across shifts.

3.2. Intended Users (Target Audience):

- LRT Duty Managers (primary users) - frontline staff who decide hourly train frequency.
- Operations Control Centre (OCC) Supervisors - oversee multiple lines and approve major adjustments.
- Shift Handover Staff - use the saved schedule and shift briefing to take over without information loss.
- Operations Planners - review historical decisions to refine standard timetables and event playbooks.
- Finance / Cost Auditors - review per-day cost deltas to verify operational spend is justified.
- Senior Management - access aggregated reports to track productivity gains and cost savings over time.

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4. System Functionalities

4.1. Description: The system operates as a hybrid decision-intelligence engine. A deterministic math layer (calibrated against actual Malaysian LRT ridership data) computes the three scheduling options: conservative, moderate, aggressive - with full metrics for each: predicted passenger load, train capacity, load factor, cost delta, and time saved per passenger. Z.AI's GLM then acts as the reasoning layer: interpreting unstructured situation input, evaluating the three options against operational policies and historical precedents, selecting the optimal choice, and explaining the trade-offs in plain language. The duty manager remains in control which they can override GLM's pick but the heavy analytical lifting is done for them.

4.2. Key Functionalities:

- **Multi-Modal Situation Capture:** Duty manager describes the situation via manual form or free-text (example: "Blackpink concert at Bukit Jalil, 4,000 extra pax/hr, rainy"). GLM extracts 3 structured operational factors - weather, event type and emergency type then adjust the affected line from any of the three input modes.
- **Three-Option Trade-Off Analysis:** For every situation, the system generates three differentiated scheduling options with full quantitative metrics (load factor, headway, cost delta, passengers served). The options are always spread around the mathematically optimal frequency so the duty manager can see the cost-service trade-off at a glance.
- **GLM-Powered Recommendation & Reasoning:** GLM reads the situation, the three options, and the most relevant historical precedents, then picks one option and streams a plain-language justification - citing load factor, headway, cost, and why the other two options are inferior. A shift briefing under 220 words is auto-generated for handover.
- **Schedule Application & Persistence:** Duty manager applies the chosen option with one click; the daily schedule updates in place with adjusted hours visually highlighted. The full updated schedule, situation parameters, and cost summary are saved to SQLite, keyed by date and line, so any past decision can be retrieved, reviewed, or used to train future operators.

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4.3. AI Model & Prompt Design

4.3.1. Model Selection: Z.AI's GLM (ilmu-glm 5.1) is used as the core reasoning model. It was selected because: (1) it handles long, structured prompts mixing tabular data with prose context (situation + three options + historical precedents in one prompt), which a smaller chat model would lose track of; (2) it supports streaming output, allowing the UI to show GLM's reasoning live to the duty manager rather than making them wait for a full response; (3) it produces consistent structured tail tokens (e.g. "RECOMMENDATION: moderate") that the system can parse reliably; and (4) Z.AI's endpoint is the official track sponsor for this hackathon, and ilmu-glm 5.1 is the Malaysia-tuned variant most likely to recognize local context ("Bukit Jalil", "Thaipusam", "pasar malam", line names) without additional fine-tuning.

4.3.2. Prompting Strategy: Multi-step structured prompting with role-based system messages. Each function (extraction, recommendation, daily briefing) has a dedicated system prompt that pins the model into the duty-manager domain and enforces output format. For situation extraction we use zero-shot JSON-only prompting with strict schema; for option selection we use a structured prompt that includes the situation, three pre-computed options with metrics, and one historical precedent (lightweight RAG ranked by line + weather + event match). We avoid few-shot examples in the recommendation prompt because the options themselves serve as the in-context grounding - the model must reason over the numbers, not pattern-match to examples. This fits our use case because every decision must be justified on the specific load factor, cost delta, and emergency type at hand, not a memorized response.

4.3.3. Context & Input Handling: Free-text situation descriptions are accepted up to ~2,000 characters; anything longer is truncated to the first 2,000 chars with a UI warning. Structured inputs (date, line, weather, event pax/hr) are validated client-side: event crowd must be 0–30,000 pax/hr, time window must be at least one hour, and event type must be selected if pax/hr > 0. Inputs that fail validation are rejected with an explicit error message instead of being silently truncated, so the duty manager always sees what was rejected and why.

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4.3.4. Fallback & Failure Behavior: Three layers of fallback. (1) If GLM streaming fails mid-response, the system retries once with a non-streaming blocking call. (2) If GLM is fully unreachable (timeout or connection error), the system falls back to a deterministic math-based recommendation ("moderate is optimal at this load factor") with an explicit 'GLM unavailable' warning shown to the user. (3) For situation extraction specifically, if GLM returns malformed JSON or off-topic content, a keyword-based regex extractor takes over, capturing weather (rain/storm/cloudy), emergency type (track/signal/power/breakdown/evacuation/overcrowding), line name, and event type from common Malay/English keywords. Output from any fallback path is clearly labelled in the UI so the duty manager knows the recommendation came from rules, not the model, and can escalate to a human supervisor if needed. Hard safety rules (e.g. track incident → zero trains, full suspension) are enforced deterministically and cannot be overridden by GLM — if GLM hallucinates a contrary recommendation, the math layer overrides it.

5. User Stories & Use Cases

- **User Stories:** "As a duty manager, I want to describe a situation in plain text so that the system can extract the factors and recommend a frequency adjustment without me looking up ridership tables manually." "As a shift handover staff, I want to see the saved schedule and a written briefing so that I can take over the shift without losing context. " "As an operations planner, I want to retrieve any past adjusted schedule by date and line so that I can audit the decision and refine future playbooks." "As a finance auditor, I want to see standard cost vs extra cost per day so that I can verify operational spend is justified by demand."
- **Use Cases (Main Interactions):**

  Situation-to-Schedule Flow: Duty manager picks the date and line, describes the situation (manual / text), and clicks Analyse. The system extracts structured factors via GLM, computes three scheduling options instantly, then GLM streams its reasoning and picks one. The manager reviews and either accepts GLM's pick or selects another option, then clicks Apply to update the day's schedule and Save to persist.

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Saved-Schedule Retrieval Flow: When the duty manager re-opens a previously adjusted date, the system loads the saved schedule from SQLite and reconstructs the full view (highlighted hours, situation parameters, cost summary). The shift briefing remains available so any incoming staff can read the rationale without rerunning GLM.

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

6. Features Included (Scope Definition)

- Two-input situation capture (manual form and free-text with GLM extraction) covering all three disruption factors: weather, emergency, and event.
- Hybrid decision engine: deterministic math layer for the three scheduling options + Z.AI GLM (ilmu-glm 5.1) reasoning layer for selection, justification, and shift briefing - with live streaming output.
- Full daily schedule view (06:00–24:00) with adjusted hours visually highlighted, plus standard / extra / total cost breakdown, weekly cost overview, and one-click Apply / Reset / Save controls.
- SQLite persistence keyed by (date, line) for full audit trail, with safety overrides (e.g. track-incident → zero trains) enforced deterministically regardless of GLM output.

7. Features Not Included (Scope Control)

- No real-time integration with live LRT operational systems (signalling, fare gates, CCTV) - the prototype is a decision-support layer that produces recommendations; physical schedule deployment is still done by OCC operators in the existing system.
- No multi-line cascade modelling - each adjustment is computed per line independently; cross-line passenger interchange impacts are not optimized in this version.
- No multi-user concurrent editing or role-based access control - the MVP assumes one duty manager per session; SSO and approval workflows are deferred to v2.

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8. Assumptions & Constraints

- **LLM Cost Constraint:** Estimated token cost per session: roughly 1,500–2,500 input tokens and 400–700 output tokens per Analyse action (situation + three options + one historical precedent in, reasoning + recommendation out). With ilmu-glm 5.1 pricing assumed at parity with similar GLM-tier models, a typical session of 2–3 Analyse actions costs well under one cent. Cost-control design decisions: (1) the math layer pre-computes the three options so GLM only has to choose, not compute - keeping the prompt tight and the output short; (2) historical precedents are limited to one record in the option prompt to bound input length; (3) shift briefings are streamed once per update and cached in session state, so reloading the page does not re-bill GLM; (4) the keyword-based extractor handles trivially obvious situations (e.g. just "rainy") without any GLM call.
- **Technical Constraints:** The MVP runs as a Streamlit web app with a SQLite backend, deployed as a single-process service. It is designed for an ops-room desktop browser, not mobile. Historical precedents are stored as JSON, not a vector database - fine for tens to low hundreds of records but would need migration for thousands.
- **Performance Constraints:** End-to-end latency target is under 30 seconds from clicking Analyse to having an applied schedule. The math layer responds in under 1 second; GLM streaming typically completes in 8-20 seconds depending on prompt length and server load. The system does not pre-compute schedules in advance; every Analyse click triggers a fresh computation, which is acceptable at duty-room scale (a few dozen actions per day per line) but would need batch pre-computation at fleet-wide scale.
- **User Input:** The system always requires a human duty manager to confirm and apply the chosen option - GLM's recommendation is never auto-applied. The manager can override GLM's pick at any time, and all decisions (including overrides) are logged. This human-in-the-loop design is deliberate: schedule changes affect public safety, so accountability must rest with a named human, not an AI model.

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9. Risks & Questions Throughout Development

- **GLM Hallucination on Safety-Critical Cases:** How do we guarantee GLM never recommends running trains during a track incident or signal failure? - Mitigation: the math layer hard-codes zero-train output for these emergency types regardless of what GLM says, and the UI displays a deterministic safety banner before GLM finishes streaming.
- **Drift in Ridership Baselines:** How do we keep the standard ridership tables current as Klang Valley ridership patterns evolve? - Mitigation: the saved-schedule database accumulates real outcomes that can be used to recalibrate the baseline tables quarterly; this is out of scope for the MVP but is a v2 priority.
- **Adoption Risk With Frontline Staff:** Will duty managers trust GLM's recommendation enough to actually use it under real time pressure? - Mitigation: the streaming reasoning is shown in plain language with explicit references to load factor, cost, and historical precedent, so the manager can verify the logic; the manager can also override the pick, preserving their authority and building trust over time.

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